

A spatially resolved hepatobiliary pharmacokinetic and pharmacodynamic framework for MASLD

Spatially resolved hepatobiliary PK/PD model

Abstract

Drug disposition in the liver is shaped by spatially organized uptake, metabolism, canalicular export, and disease-dependent adaptation. We develop a lobule-scale computational pharmacokinetic/pharmacodynamic framework that uses human spatial multi-omics to represent portal, periportal, mid, and central liver states across control, MASL, and MASH conditions. The model includes sinusoidal uptake through SLC10A1, SLCO1B1, and SLCO1B3; canalicular export through ABCB11, ABCC2, and ABCB4; and compensatory basolateral routes through ABCC3 and ABCC4. We use gadoxetate as a clinically anchored tracer for uptake and hepatobiliary clearance, and rosuvastatin as a transporter-mediated drug probe for exposure and DDI analysis. A counterfactual analysis using identical mean transporter capacities shows that spatial zonation changes predicted MASH bile exposure by approximately 15% relative to a well-mixed liver and changes hepatocyte exposure by approximately 10%. A pilot sensitivity analysis also predicts increased injury under reduced export and obstruction-like stress. The framework is designed to answer pharmacology questions: how disease changes exposure, which transporters control clearance, what perturbations increase DDI risk, and how exposure may be linked to efficacy or toxicity. The current results are exploratory and require compound-specific calibration before clinical prediction.

Keywords: hepatobiliary pharmacokinetics; pharmacodynamics; MASLD; liver zonation; transporter-mediated DDI; gadoxetate; rosuvastatin

1 Introduction

Many liver PK models treat the organ as homogeneous even though hepatocytes occupy distinct microenvironments along the portal-central axis. Spatial organization affects oxygen exposure, substrate availability, metabolic capacity, bile-acid handling, and susceptibility to injury. This is especially relevant in MASLD, where steatosis, inflammation, fibrosis, and transporter adaptation can change both efficacy-relevant exposure and safety margins.

The 2025 Nature Genetics human MASLD atlas provides a current spatial reference for this problem. It combines single-cell transcriptomics, spatial transcriptomics, spatial metabolomics, and spatial proteomics across 61 individuals, including control, MASL, and MASH cohorts [1]. The study orders spatial transcriptomic spots into portal, periportal, mid, and central zones and identifies disease-associated spatial metabolic programs. A complementary 2025 Nature study uses spatial metabolic gradients and bile-acid signals to define portal-to-central metabolic organization [2].

The pharmacological opportunity is to use these spatial states as model coordinates rather than as descriptive labels. A zonated model can propagate a compound through uptake, intracellular effect or metabolism, canalicular export, and compensatory basolateral clearance. It can then report exposure, effect, toxicity, and DDI endpoints that are directly relevant to drug development.

This proposal extends, rather than replaces, the state of the art in transporter PBPK. Rosuvastatin PBPK models have already been qualified against large clinical datasets, including plasma, urine, feces, PET liver measurements, and multiple clinical DDI studies [9]. Their strength is quantitative whole-body prediction; their usual limitation is that hepatic transporter capacity is represented as a lumped organ-level quantity. The unresolved question addressed here is whether the spatial placement of that capacity changes disease- and inhibitor-dependent predictions even when the organ-level mean is unchanged.

This paper therefore focuses on a pharmacology-facing framework. Anatomical network visualization and macroscopic biliary-tree reconstruction are outside its scope. The model is intentionally organized around compound disposition, transporter mechanisms, disease-state perturbations, and measurable PK/PD outputs.

2 Materials and methods

2.1 Spatial disease-state representation

The model uses four lobular zones: portal, periportal, mid, and central. Disease states are control, MASL, and MASH, matching the human spatial atlas [1]. Each zone carries relative capacities for uptake, canalicular export, basolateral escape, metabolism, and injury susceptibility. The capacities are dimensionless until replaced by normalized expression, proteomic, or functional measurements.

2.2 Transporter mechanisms

Mechanism	Genes	Pharmacological interpretation
Hepatic uptake	SLC10A1, SLCO1B1, SLCO1B3	Controls entry from portal blood and hepatic extraction
Canalicular export	ABCB11, ABCC2, ABCB4	Controls delivery to bile and biliary clearance
Basolateral escape	ABCC3, ABCC4	Alternative blood-facing clearance during cholestatic stress
Cholangiocyte modification	CFTR, SLC4A2	Optional ductular secretion module; not used in the present PK core

The core equations are a reduced mass-balance system. Parent compound enters the hepatocyte according to zoned uptake capacity, is transformed or acts on an intracellular target, and is cleared through canalicular and basolateral routes. DDI simulations alter transporter activity using an inhibition factor, while disease simulations alter the zonal capacity fields. This formulation allows exposure ratios, clearance changes, and effect-site differences to be compared without requiring a macroscopic biliary geometry.

2.3 Reduced mass-balance model

For each lobular zone i , the model tracks parent compound in plasma, hepatocytes, and bile. The reduced equations are written below in rendered form; all rate constants are non-negative and may depend on disease state or transporter perturbation.

$$\begin{aligned}dA_{hi}/dt &= k_{ui} A_p - (k_{ci} + k_{\beta i} + k_{mi}) A_{hi} \\dA_{\beta}/dt &= \sum_i k_{ci} A_{hi} - k_l A_{\beta} \\E_i &= E_{max} C_{hi} / (EC_{50} + C_{hi})\end{aligned}$$

Here A_p is the circulating parent amount, A_{hi} is the hepatocyte amount in zone i , A_{β} is the bile amount, k_u is uptake, k_c is canalicular export, k_{β} is basolateral escape, k_m is intracellular metabolism or irreversible loss, and k_l is bile-side loss. C_{hi} is the corresponding intracellular concentration. Transporter inhibition multiplies the affected rate constant by an activity factor between zero and one.

2.4 PK and PD endpoints

The primary PK endpoints are plasma Cmax, area under the concentration-time curve, time to maximum concentration, hepatic exposure, biliary exposure, and route-specific recovery. The principal DDI endpoint is the exposure ratio relative to the victim-drug control condition. For a therapeutic compound, efficacy can be represented with an effect-compartment or Emax relationship, in which effect increases with the active intracellular concentration and is bounded by a compound-specific maximum. Toxicity is represented separately from efficacy using a normalized injury state linked to intracellular and bile-side burden.

2.5 Gadoxetate reference calibration

Gadoxetate is used as a tracer calibration compound because it has a clinically characterized hepatobiliary disposition. The recommended dose is 0.025 mmol/kg intravenously. The product label reports dose-linear kinetics up to 0.1 mmol/kg, a distribution volume of approximately 0.21 L/kg, plasma protein binding below 10%, a terminal half-life of approximately 0.91-0.95 h, total clearance of about 250 mL/min, and renal clearance of about 120 mL/min [5]. In subjects with normal liver and kidney function, approximately half of the dose is eliminated hepatobiliary and half renally. Severe hepatic impairment can reduce hepatobiliary excretion to approximately 5% [5].

Human dynamic MRI studies support a two-compartment uptake and efflux interpretation, with aggregate uptake reflecting OATP1B1, OATP1B3, and NTCP contributions and intracellular efflux reflecting canalicular export dominated by MRP2 [6]. These values provide external checks on the model's dose, terminal kinetics, and route-of-excretion fractions.

2.6 Zonation counterfactual

To test whether spatial structure changes pharmacology-facing predictions, we compare the zoned model with a well-mixed counterfactual. The two models receive the same input pulse, use the same mean transporter capacities within each disease state, and differ only in whether those capacities are distributed across portal, periportal, mid, and central positions. We evaluate baseline exposure, 50% uptake inhibition, and 65% MRP2 inhibition. This isolates the contribution of spatial organization from the contribution of total clearance capacity.

3 Results

3.1 Zonation changes pharmacology-facing predictions

The central counterfactual result is that spatial arrangement changes predicted exposure even when the mean transporter capacity is held constant. In the MASL baseline, bile AUC was 3071.28 normalized units in the zoned model versus 3277.67 in the well-mixed model, a 6.3% difference. In MASH, bile AUC was 2540.11 versus 2983.36, a 14.9% difference, while hepatocyte parent AUC was 1.137 versus 1.036, a 9.8% increase in the zoned model. Under 65% MRP2 inhibition, the MASH bile-AUC difference increased to 17.2%. The effect is therefore largest when disease and transporter perturbation are combined.

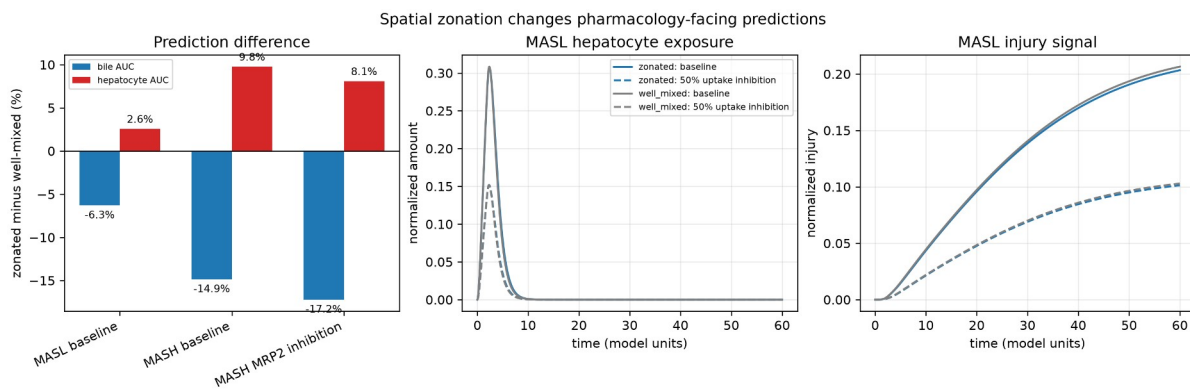


Figure 1. Counterfactual test of spatial zonation. Blue curves use the zoned model; gray curves use a well-mixed model with identical mean capacities. Solid lines are baseline and dashed lines represent 50% uptake inhibition.

3.2 Atlas-informed transporter sensitivity

The atlas-informed pilot model predicts that disease-state changes in uptake and export can alter exposure even when the nominal input pulse is unchanged. In the current normalized runs, the control baseline produced a bile-associated Cmax of 21.92 and an exposure AUC of 190.16 model units. The MASL baseline produced a Cmax of 86.76 and an AUC of 3071.28, while the MASH baseline produced a Cmax of 69.96 and an AUC of 2540.11. These values are not clinical concentrations; they are sensitivity outputs showing the effect of zoned capacity assumptions.

Reducing MRP2-like export from 1.0 to 0.35 increased the control peak injury signal from 0.0124 to 0.0211. A 50% obstruction increased control exposure AUC from 190.16 to 380.52 and peak injury from 0.0124 to 0.0234. These results are sensitivity outputs; the disease-state comparison should not be overinterpreted until transporter capacities are replaced with measured expression or functional data.

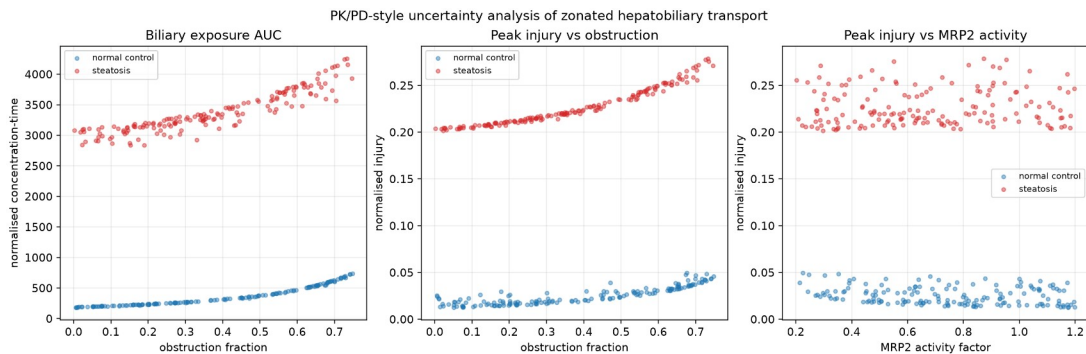


Figure 2. Exposure and injury sensitivity to obstruction-like stress and MRP2 activity. The outputs are normalized and intended for model qualification, not clinical dosing.

3.3 Gadoxetate PK calibration scaffold

The gadoxetate scaffold reproduced a 24-hour hepatobiliary recovery of 50.1% in the healthy reference condition, matching the approximate label-level 50:50 renal/hepatobiliary split. The exploratory MASL/MASH condition recovered 37.6% through the modeled hepatobiliary route, while the severe hepatic-impairment condition recovered 4.8%, consistent with the label's approximately 5% value. This gives the model a quantitative calibration target before applying it to a therapeutic drug.

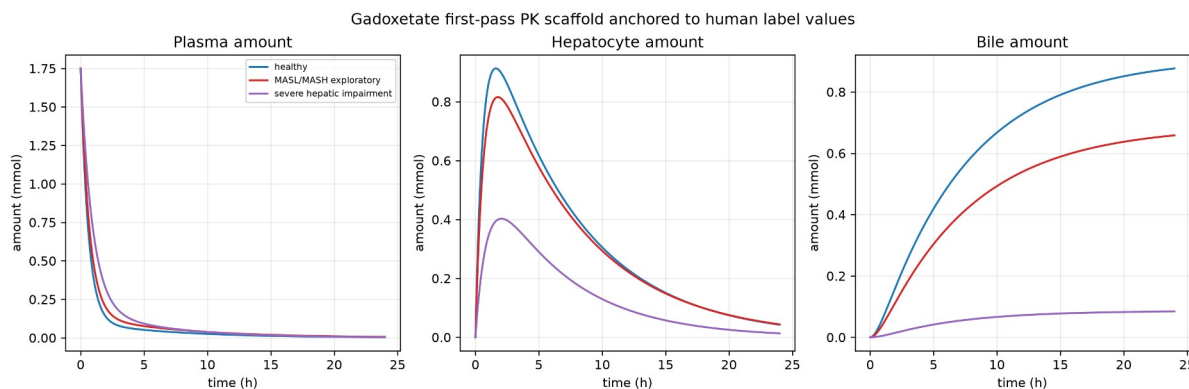


Figure 3. Gadoxetate plasma, hepatocyte, and bile amounts under healthy, exploratory MASL/MASH, and severe hepatic-impairment scenarios.

3.4 DDI and efficacy use case

Rosuvastatin is a suitable next drug probe because its disposition depends on hepatic uptake and efflux transporters, it undergoes limited metabolism, and it has human transporter and PET literature [7]. In the framework, an interacting perpetrator can inhibit SLCO1B1/SLCO1B3, ABCG2, ABCC2, or ABCC4. The model then reports victim-drug AUC and Cmax ratios, hepatic concentration changes, and changes in bile-associated exposure. These outputs map directly to early DDI risk assessment.

For efficacy, a therapeutic compound can be assigned a target-specific effect compartment and an exposure-response relationship. The relevant comparison is not simply whether biliary concentration rises, but whether active intracellular exposure crosses the effect threshold while systemic and hepatic exposure remain below toxicity limits. This separation is the principal reason to retain the zoned lobule model for a pharmacology paper.

4 Discussion

This second paper is centered on a drug-development question: how spatial liver biology changes compound exposure, effect, and DDI risk. The key novelty is the counterfactual demonstration that a zoned liver can produce different drug-disposition predictions from a well-mixed liver even when mean transporter capacity is identical. The Nature Genetics atlas supplies human disease-state organization; the 2025 spatial-gradient study supplies metabolic context; and gadoxetate provides a clinically characterized tracer for model qualification.

The model also makes a useful mechanistic distinction between uptake limitation and export limitation. Lower OATP-mediated uptake can reduce hepatic and biliary exposure while increasing plasma exposure. Lower MRP2-like export can increase intracellular retention and injury risk, while increased ABCC3/ABCC4-like escape can partially restore blood-facing clearance. These mechanisms generate different PK signatures and therefore different DDI and safety interpretations.

The primary limitations are quantitative. The current zonal transporter fields are hypothesis parameters rather than direct atlas-derived fold changes. The disease states are not yet linked to individual patient covariates, and the therapeutic efficacy module requires a compound-specific target and clinical exposure-response data. Gadaxetate is a calibration tracer, not a surrogate for every drug substrate. Finally, transporter expression is not identical to functional transport capacity, so expression-based updates should be checked against hepatocyte experiments, clinical DDI studies, or imaging-derived uptake rates.

The next development stage is a compound-centered benchmark. Rosuvastatin can test transporter-mediated disposition and DDI behavior, followed by a therapeutic candidate with a defined efficacy biomarker and exposure-response relationship. Success should be judged by prediction of plasma exposure, hepatic uptake, biliary recovery, and DDI direction across control and MASLD states, rather than by visual fidelity of an anatomical tree.

5 Conclusions

We present a spatially resolved hepatobiliary PK/PD framework designed for drug efficacy, safety, and DDI questions. Its testable novelty is that disease- and transporter-dependent spatial arrangement changes pharmacology-facing predictions beyond what is captured by a well-mixed liver with the same mean capacity. The framework uses a human four-zone MASLD atlas, explicit transporter mechanisms, and gadoxetate PK data for calibration. It is intentionally independent of macroscopic biliary-tree visualization and is structured for future compound-specific validation.

Data and code availability

The model code and normalized scenario outputs are maintained with the project materials. The Nature Genetics processed spatial datasets are available through the Human MASLD Spatial Multiomics Atlas and the study accessions HRA007511, OMIX009098, OMIX010136, and OMIX009117. The current transporter coefficients are transparent hypothesis parameters and will be replaced by atlas-derived or experimentally measured values in the next calibration release.

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